

(Action Engine)

1. Priority → Action mapping — built on real Ahmedabad HAP, not invented

Researched Ahmedabad's own, already-implemented Heat Action Plan (HAP) — the city has run this since 2013, South Asia's first municipal HAP. Its real color-coded alert system (Yellow 41.1–43°C / Orange 43.1–44.9°C / Red >45°C) scales actions from awareness → infrastructure readiness → cooling-center activation → hospital/resource mobilization as conditions worsen. Our tier ladder (LOW/MODERATE/HIGH/CRITICAL) mirrors that same real escalation pattern — not the literal temperature thresholds (our tiers are multi-day CTBI-based, not single-day temperature), but the same increasing-intensity logic.

Our tier	Mirrors HAP's	Core actions
LOW	Yellow-day baseline	Awareness pamphlets, ensure water points functional
MODERATE	Yellow→Orange transition	Fans/shade at transit hubs, brief health workers, pre-position water/ORS
HIGH	Orange/Red	Activate cooling centers, shift outdoor work hours, alert hospitals
CRITICAL	Red + resource crunch	Full cooling-center activation, mobile medical teams, prioritize ward for resource allocation , community-leader outreach (not SMS-only — per HAP's own finding that word-of-mouth reaches low-literacy groups better)

2. Driver-specific add-ons (targeted, not generic)

On top of the base tier action, wards get an EXTRA action only for the specific reason they're flagged vulnerable:

- **High population density** → crowd-accessible cooling centers + mist sprinklers at high-footfall junctions
- **Low NDVI (green cover)** → portable shade structures as an interim measure

This keeps recommendations targeted — a ward flagged for density gets a different add-on than a ward flagged for lack of greenery, even at the same priority tier.

3. QA — dashboard/action consistency check (done)

Verified programmatically across all 48 wards (latest-date snapshot):

- Every HIGH-tier ward's actions include cooling-center activation 
- Every LOW-tier ward's actions do NOT include cooling-center activation (no over-triggering) 
- Every CRITICAL-tier ward's actions include resource-prioritization language 
- **0 inconsistencies found** — tier and recommendation always match.

4. What this is NOT

- No medical treatment instructions anywhere (matches playbook: "public-health preparedness recommendations", not medical prescriptions)
- No CRITICAL tier reachable yet in practice — same honest cap as priority_rubric_v2 (D10): CRITICAL requires real response-capacity data, which is still pending (healthcare-access/OSM not built). Action engine correctly inherits this cap — it will only ever recommend CRITICAL-tier actions once the rubric itself can justify a CRITICAL verdict.

Files produced

File	Location
khushi_day4_action_engine.py	src/
action_engine_full_test.csv (26,208 rows)	data/processed/
action_engine_test_snapshot.csv (48 wards)	data/processed/
Khushi_Day4_Completion_Summary.md (this file)	docs/

Sources (Ahmedabad HAP, paraphrased)

- Bloomberg, "An Indian City's Fight Against Deadly Heat Holds Lessons for Others" (2023)
- "Cool Roofs, Hot City: Ahmedabad's Battle to Adapt to Extreme Urban Heat" (2024)
- ClimaHealth, "Ahmedabad's Heat Action Plan: Development and lessons learned"

